# Beyond Accuracy: Diagnosing Analogical Transfer in Language Models

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## Abstract

An accuracy gain after an analogy does not establish that a language model used its relational structure, and an incorrect answer does not identify where transfer failed. We introduce a theory-grounded diagnostic framework with explicit artifacts for source construction, mapping, schema induction, and adaptation. Restricted interfaces make the source-to-target path inspectable; oracle-authored replacements allow the same base model to continue from corrected intermediate guidance. Three studies examine outcome attribution, source-answer transfer, and stage diagnosis. An audit of 400 natural model–item outcomes finds 12 correct answers despite absent schema content at adaptation. Scientific question-answering controls show that net gains can coexist with regressions and need not exceed explanation or random-example benefits. A four-condition narrative study finds positive structure-dependent source-answer interactions in four of six models, robust in conditional story-level resampling and control sensitivity analyses. In a 200-question Llama diagnostic study, upstream replacements rescue 18 of 73 initial GPQA errors and 15 of 43 murder-mystery errors, while many corrected continuations remain unsuccessful. These observations distinguish first-artifact assignments from subsequent continuation outcomes. The framework provides an auditable evaluation protocol; its artifacts do not reveal hidden cognition, and assisted continuations are distinct from autonomous model performance.

## 1 Introduction

A solved example can help a model answer a new question, introduce a misleading rule, or supply information that the model never uses. These possibilities matter for analogy-based assistance: its intended benefit comes from transferring a useful relation across situations, rather than merely adding text or copying an answer. Yet target accuracy alone does not distinguish them. In our scientific question-answering study, a 300-word teacher analogy changes 86 Mistral answers from incorrect to correct and 41 from correct to incorrect, relative to chain-of-thought (CoT), on the same 448 questions. The net gain hides substantial regressions, and a non-analogical explanation achieves slightly higher accuracy than the analogy.

Analogical prompting has shown that language models can benefit from self-generated exemplars (Yasunaga et al., 2024), while teacher analogies can support scientific concept understanding (Yuan et al., 2024). However, randomly selected examples can also help (Qin et al., 2025). A contrast against one baseline therefore leaves competing explanations: added information, a changed reasoning instruction, a source-answer cue, or transfer through the intended correspondence. Even when a source is appropriate, an unsuccessful answer leaves a further question: was the source poorly chosen, its relation lost, its schema inadequate, or its application incorrect?

We connect three evaluation questions: *what changed in the answer, does the benefit depend on the designed correspondence, and what deficiencies occur along an explicit transfer attempt?* These require different evidence. Matched correctness transitions describe improvements and regressions. Controlled source assignment and answer availability test selective source benefits. Intermediate artifacts and continuation tests characterize an observable source-to-target path.

Our central contribution is a **theory-grounded diagnostic framework** operationalizing source construction, mapping, schema induction, and adaptation. It translates established cognitive distinctions into recorded outputs with explicit information contracts. Mapping compares solved sources; schema induction abstracts their common structure; adaptation reconnects that structure to the target. We inspect unsuccessful paths and re-

place inadequate artifacts with oracle-authored references before asking the same base model to continue. The resulting record separates the first assigned deficiency from the outcome and deficiencies of subsequent corrected continuations.

Three studies support this account. Scientific question-answering controls motivate careful attribution. A factorial narrative experiment provides positive evidence of selective source-answer benefit in several models. A diagnostic study then examines the imposed four-stage workflow. Its contribution is diagnostic resolution rather than an accuracy advantage: the direct and staged baselines are not compute-matched, and supplying a reference-informed artifact is external assistance. The studies use different cohorts and protocols; their relationship is conceptual rather than a causal comparison of supplied versus self-generated sources.

We contribute (1) an executable, artifact-based operationalization of theoretical distinctions in analogy; (2) an evaluation design separating outcome changes, structural selectivity, and stage-level continuation outcomes; and (3) empirical analyses with paired accounting, conditional uncertainty estimates, and explicit separation of natural performance from oracle assistance. The framework makes a designed process inspectable without claiming that language models internally reason in four discrete stages.

## 2 A Theory-Grounded Diagnostic Framework

### 2.1 Theory translated into information contracts

Structure-mapping theory distinguishes relations between systems from resemblance between their attributes (Gentner, 1983). Systematicity favors connected relations over isolated local matches (Clement and Gentner, 1991), while computational structure mapping makes consistency constraints explicit (Falkenhainer et al., 1989). Access to a source is separable from the soundness of an inference drawn from it (Gentner et al., 1993). Comparative encoding can support schema induction (Gick and Holyoak, 1983; Gentner et al., 2003), and applying that schema to new constraints remains a further challenge (Novick and Holyoak, 1991). These distinctions motivate separately inspectable outputs. They do not prescribe a unique linear architecture.

Let $x$ denote a target with choices, $r$ solved sources, $m$ their comparative mapping, $s$ a schema, and $a$ the target solution. Our permitted flow is

$$\begin{aligned} r &= R(x), \quad m = M(r), \\ s &= S(m), \quad a = A(x, s). \end{aligned} \tag{1}$$

Each stage is a fresh call without hidden conversational history. Only the designated artifact passes forward. The target is available to $R$ and $A$, but not to $M$ or $S$. The reference answer is unavailable throughout the natural workflow.

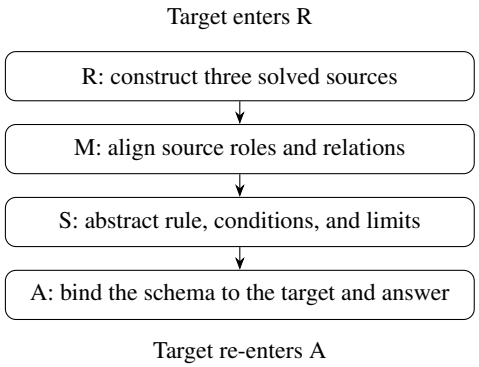



Figure 1: The imposed transfer path. Middle stages compare and abstract sources without the target. Adaptation receives the schema and target without raw source narratives.

**Source construction (R).** The historical stage name is *retrieval*; operationally, the model generates three solved problems. We observe those sources, not retrieval from internal memory. A source can resemble the target topically yet omit the relation needed for a solution. An incorrect source solution can also contaminate every later artifact.

**Comparative mapping (M).** Mapping aligns the generated sources with one another, recording role correspondences, common relations, suitability, and important differences. It is not a direct source-to-target mapping because the target is withheld. The final connection to the target occurs in $A$. Connected relations and their constraints matter more than renamed entities (Gentner and Markman, 1997).

**Schema induction (S).** The model sees the mapping artifact rather than the raw narratives. Its output states a solution rule and when that rule is applicable. A slogan such as "compare the evidence" does not specify how evidence distinguishes alternatives; an over-specific rule may merely restate one source. The schema must preserve useful

| Stage | Permitted input | Required content and grounds for inspecting adequacy |
|---|---|---|
| R: source construction | Target and choices | Three solved source problems, answers, and explanations. Inspect source-solution correctness and whether the relations can support the target. |
| M: mapping | Recorded R artifact | Shared roles and relations, consequential differences, and source suitability. Inspect consistency and preservation of the solution-relevant relation. |
| S: schema induction | Recorded M artifact | Abstract rule, roles, relations, applicability conditions, and exceptions. Inspect whether abstraction retains sufficient constraints and operations. |
| A: adaptation | Target and schema | Target bindings, applicability checks, adjustments, and final answer. Inspect whether the rule applies and whether the answer follows from target evidence. |

Table 1: Information and content contracts. These are semantic inspection criteria, not a validated automatic classifier or a newly measured annotation-agreement score.

operations while identifying boundaries of generalization.

**Adaptation (A).** The model instantiates schema roles in the target, checks constraints, makes necessary adjustments, and answers. Correctness and schema use are separate: a solver may reject a schema and solve independently, or report using it while making an invalid inference. Output fields and self-reports are observable traces rather than proof of faithful reasoning.

As an illustrative theoretical example, the army/radiation analogy preserves a convergence relation: strong force along one route is unsafe, weak forces alone are insufficient, and coordinated weak forces combine at a goal (Gick and Holyoak, 1980). Source construction must supply that relation; mapping aligns its roles; a schema preserves its conditions; adaptation checks whether the target satisfies them. This illustration is not one of our benchmark observations.

The interfaces deliberately test a source-comparison-to-schema-to-target route. Withholding information makes the support for each call traceable, but may impose bottlenecks absent from ordinary prompting. We therefore evaluate behavior under these contracts rather than identifying a universal sequence of cognitive operations.

### 2.2 Inspection and cumulative continuation

On a naturally unsuccessful path, artifacts are inspected in stage order and the first inadequate output is assigned a stage label. This ordering distinguishes upstream deficiencies from later ones without making the label a unique causal diagnosis. A malformed artifact and a semantically inadequate artifact also require different explanations; valid JSON alone is insufficient.

We replace the assigned artifact, keep earlier artifacts, and rerun downstream stages with the same base model. A correct final answer counts as an *oracle-assisted continuation success*. If the answer remains incorrect, the new path is inspected and a further reference replaces its first inadequate artifact, retaining earlier corrections. The protocol records an initial assignment and a trajectory of corrected continuations. A question may contribute to several stage burdens, so those burdens are not a partition of questions.

In the primary study, GPT-6 Sol authors reference artifacts with access to the target answer: *open-oracle authoring*. Upstream references are intended to respect stage-content requirements and avoid explicit disclosure of the target answer. The continuing base model receives the artifact rather than the key. Nevertheless, a reference can indirectly carry solution guidance, and its quality is not guaranteed simply because a stronger model produced it.

These interventions measure response to a particular corrected input under an accumulated history. They do not isolate the effect of correcting a stage from changes in wording, information, or stochastic continuation. A subsequent deficiency is observed in a new continuation; we cannot prove it already existed in the unexecuted counterfactual continuation of the original path.

Replacing $A$ directly supplies the final answer and leaves no downstream base-model call. Such outputs are excluded from continuation-success counts. This distinction separates correct oracle answers from successful use of upstream guidance by the original solver.

## 3 Evaluation Design

We study three questions: **RQ1**, how do analogy conditions compare with alternative assistance and what answer changes accompany them; **RQ2**, does a source answer help more under intended correspondence than random assignment; and **RQ3**, what do explicit artifacts and corrected continua-

tions reveal about unsuccessful staged paths?

### 3.1 Tasks and accounting

GPQA Main provides 448 graduate-level questions (Rein et al., 2024): 78 biology, 183 chemistry, and 187 physics items in our local version. We use Main rather than Diamond. MuSR provides murder mysteries, object-placement questions, and team-allocation problems requiring multistep reasoning over narratives (Sprague et al., 2024).

Accuracy uses the benchmark choice key. Recorded responses without a valid final answer count as incorrect. Missing records are distinct from recorded invalid answers. Matched comparisons align question identifiers. Repeated executions are kept with their question rather than counted as independent examples. Different tasks, samples, prompts, and providers preclude interpreting cross-study contrasts as controlled effects.

### 3.2 RQ1: outcome attribution and transitions

DeepSeek V4 Flash supplies concept-focused teacher analogies and non-analogical explanations for GPQA. Solvers are Llama-3.1-8B-Instruct, Qwen3-8B, Mistral-7B-Instruct-v0.3, and Gemma-4-31B-it. Conditions include 300- and 600-word analogies; two 300-word or three 200-word examples; 300- and 600-word explanations; random same-domain and other-domain examples; CoT; and solver-generated analogy (Wei et al., 2022; Kojima et al., 2022). Word budgets are nominal, not compute matching. Appendix A defines all conditions.

On shared items, let $H_{b\to c}$ be the fraction changing from incorrect under baseline $b$ to correct under condition $c$, and $D_{b\to c}$ the reverse. Then

$$p_c - p_b = H_{b\to c} - D_{b\to c}. \tag{2}$$

Both fractions use all shared items as denominator. We also screen explicit references to the analogy, example, or teacher, including rejection. This lexical observation is separate from semantic transfer validation.

Llama, Qwen, and Mistral have 448 retained responses per condition. Gemma has 435–447, so its full-cohort accuracies can reflect composition differences. We do not interpret small unmatched Gemma differences as isolated treatment effects.

### 3.3 RQ2: source type by answer availability

GPT-5.6-Sol constructs a source with changed entities and context but intended shared solution relations for each selected MuSR target. It sees the target key; the solver does not. A source contains a narrative, question, choices, and source answer. Difficult examples receive manual review and a small model-based assessment; there is no exhaustive independent structural annotation.

The completed set comprises 103 questions: 61 murder mysteries, 21 object placements, and 21 team allocations, spanning 88 target stories. It is not a documented representative random sample of MuSR. Random controls permute the same source corpus one-to-one among targets with equal choice counts and no self-assignment. Same-story and same-task matches are reduced where feasible; four five-choice object-placement controls remain within a story.

The four cells are corresponding/answer withheld ($A$), corresponding/answer provided ($B$), random/withheld ($C$), and random/provided ($D$). Withholding removes only the answer field, retaining source choices. The target, neutral "additional example" label, instruction, temperature zero, and 2,048-token ceiling are fixed. We measure

$$\Delta = (p_B - p_A) - (p_D - p_C). \tag{3}$$

A positive interaction indicates a larger source-answer benefit under the designed correspondence. It does not validate every individual transfer, and can combine improvement in $B$ with deterioration in $D$.

Llama-3.2-1B and 3B and Llama-3.1-8B and 70B have three runs per cell; Qwen3-14B and 32B have two. Provider, version, and quantization differences prevent isolating parameter count. We average each question's executions before computing the contrast. Conditional 95% resampling intervals use 20,000 target-story bootstrap samples, retaining all four cells and executions together (seed 20261012). They summarize target-story resampling with sources and assignment fixed; they do not quantify source-generation or random-permutation uncertainty.

Sensitivity analyses remove the four same-story controls and use the subset of questions with valid answers in every cell and execution. The latter conditions on an observed outcome and is a diagnostic check, not an unbiased replacement for full-cohort accuracy. We also compare matching and differing source/target answer positions; these strata contain different questions rather than randomized reorderings of the same target.

### 3.4 RQ3: natural paths and assisted continuations

The primary diagnostic cohort uses Llama-3.1-8B-Instruct on 100 GPQA and 100 MuSR murder-mystery questions. Three sources are requested per target. Unsuccessful paths undergo the cumulative procedure in Section 2.2. We report natural accuracy, initial assignments, continuation successes, and remaining deficiencies with explicit denominators.

Retained 100-item reports also permit question-matched direct/staged comparisons for Llama and DeepSeek V4 Flash on the same two cohorts. These provide performance context rather than a compute-matched superiority test. We also audit all 400 natural recorded paths for explicit input keys and for the schema content actually passed to adaptation, including paths with correct answers. Recorded downstream inputs are checked against their parent artifacts. This mechanical check does not validate relational adequacy. The full Llama stage trajectories are retained aggregate diagnostic records, not a new row-level reannotation. The four recorded stage templates share version `1.1-dev-6`; M includes suspect/motive examples even in GPQA, a possible domain bias. Some archived figures doubled counts for display; we undo that transformation and use the original 100-question cohorts.

Human review covers 30% of the primary study's questions and associated outputs for coherence and correctness. The rest have no human review, and inter-annotator agreement is unavailable. We do not present stage assignments as validated ground truth. An additional saved DeepSeek schema-replacement case illustrates artifact inspection with a recoverable individual continuation; it is not a complete independently validated replication of all diagnostic trajectories.

## 4 Results

### 4.1 RQ1: a net gain leaves attribution unresolved

The concept-focused 300-word analogy does not exceed both random controls for any of the four solvers in the retained full-cohort results (Table 4 in Appendix A). Llama performs slightly better with a same-domain random source; Qwen matches that control and trails the other-domain control; Mistral trails both. Gemma lies between its random controls, with the additional composition caveat.

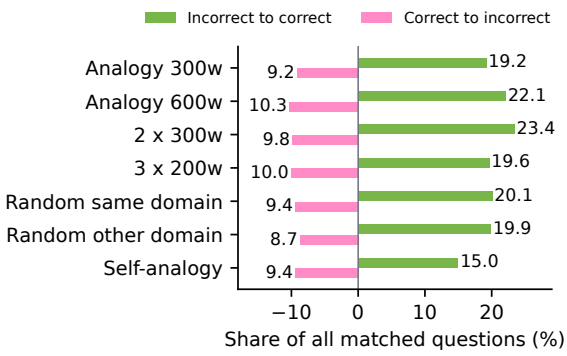



Figure 2: Mistral correctness transitions against CoT. Both directions use all 448 shared questions as denominator. Random controls can also improve or damage answers.

This does not show that relation is irrelevant in every item; it shows that accuracy here does not establish a general benefit of the intended relation.

**The baseline changes the interpretation.** Mistral improves from 16.96% with CoT to 27.01% with the 300-word teacher analogy, but a 300-word non-analogical explanation reaches 27.68%. A gain against CoT is therefore insufficient evidence that analogical presentation supplies the advantage. For Qwen, self-analogy reaches 43.30%, compared with 42.19% for CoT and 36.61% for the teacher analogy. The conditions differ in the assistance and requested procedure; they are not interchangeable measures of one capacity.

**Improvements coexist with regressions.** Mistral's teacher analogy repairs 86 CoT errors and reverses 41 correct answers: 19.20% and 9.15% of the 448 shared questions. Their difference is a 45-question gain (10.04 percentage points). Two 300-word examples repair 105 answers and damage 44. Figure 2 includes random controls, which also change both kinds of answer. These transitions are condition-associated outcomes; they do not establish a faulty analogy as the cause of each regression.

**Mention does not resolve the ambiguity.** Qwen refers to the 300-word example in 31.70% of all responses. The rate is 31.10% within correct answers and 32.04% within incorrect ones. Mention can accompany misuse or rejection, while absent mention does not establish non-use. Similarly, increasing example count creates an opportunity for schema induction rather than direct evidence of it. These observations motivate evaluating correspondence

| Model | $A$ | $B$ | $C$ | $D$ | $\Delta$ | 95% interval |
|---|---|---|---|---|---|---|
| Llama-1B | 38.51 | 44.01 | 38.51 | 44.01 | 0.00 | $[-15.46, 15.00]$ |
| Llama-3B | 54.05 | 56.96 | 40.13 | 43.04 | 0.00 | $[-9.67, 9.33]$ |
| Llama-8B | 52.10 | 65.70 | 52.43 | 51.78 | 14.24 | [5.84, 22.34] |
| Llama-70B | 59.22 | 72.17 | 53.07 | 53.07 | 12.94 | [5.35, 21.28] |
| Qwen-14B | 66.02 | 75.24 | 60.68 | 60.68 | 9.22 | [1.94, 17.00] |
| Qwen-32B | 68.93 | 74.27 | 65.53 | 58.25 | 12.62 | [1.50, 23.79] |

Table 2: Mean accuracy (%) on the same 103 MuSR questions; $\Delta$ and interval endpoints are percentage points. Intervals resample 88 target-story groups while retaining each question's four cells and all executions. They are conditional on the fixed source corpus, assignment, and prompts; no multiplicity-adjusted or source-regeneration inference is claimed.

and explicit intermediate content.

### 4.2 RQ2: selective source-answer benefit is observable

Table 2 reports every cell and the interaction. Four models show positive interactions: Llama-8B, Llama-70B, Qwen-14B, and Qwen-32B. Their conditional story-resampling intervals remain above zero. The 1B and 3B models have zero mean interaction, despite different baseline accuracies and potentially useful corresponding source content. We interpret the pattern within the retained design, without a universal scaling or hidden-cognition claim.

The four-cell view matters. Llama-8B gains 13.59 points when a corresponding source answer is added, while the random-answer contrast is $-0.65$. Qwen-32B gains 5.34 points under correspondence and loses 7.28 under random assignment: its interaction is not wholly improvement from useful guidance. A positive interaction distinguishes conditions, not necessarily a uniformly beneficial intervention.

**Control and extraction sensitivity.** Removing the four same-story controls leaves 99 questions. Interactions remain positive at 13.80, 13.13, 9.60, and 12.63 points for Llama-8B, 70B, Qwen-14B, and 32B. Restricting to questions with valid outputs in every saved cell/run also leaves positive interactions: 14.05, 14.02, 12.07, and 10.00, on 102, 88, 58, and 65 questions. The substantial Qwen invalid-output counts make this check relevant. Its selected subsets cannot replace the primary results; Appendix A.3 reports all models and counts.

**Answer positions provide a competing explanation.** With a corresponding source answer, Llama-1B achieves 88.24% when source and target answer positions match and 0.64% when they differ, pooling three runs. This is consistent with predominant index following. Llama-8B and 70B retain 57.05% and 65.38% in the differing-position stratum, compared with 74.51% and 79.08% when positions match. The larger models' behavior is compatible with more than index copying, although position sensitivity persists. Different questions form the two strata, so this is supporting evidence rather than a controlled positional intervention.

| Model | Task | Direct | Staged |
|---|---|---|---|
| Llama-8B | GPQA | 25 | 27 |
| Llama-8B | Murder | 63 | 57 |
| DeepSeek | GPQA | 58 | 63 |
| DeepSeek | Murder | 78 | 77 |

Table 3: Correct answers among the same 100 questions in each row. Both models use the same item sets. These natural results contain no oracle replacement; the direct/staged comparison is not compute-matched.

Taken together, these results support selective benefit under an intended source relation in several evaluated models. They do not establish that every correct answer used a valid mapping, nor that the same model can construct and abstract adequate sources autonomously. The latter distinction motivates inspecting the staged path.

### 4.3 RQ3: initial assignments and continuations differ

**Natural performance gives context.** Table 3 compares direct and staged answers on matched 100-item sets. Llama's staged workflow answers 27 GPQA and 57 murder questions correctly. It improves over direct answering on GPQA but declines on murder mysteries. DeepSeek shows the same direction of change, with smaller murder decline. Multiple calls and retries are not matched to the direct baseline, so these results establish neither efficiency nor an accuracy advantage of the architecture.

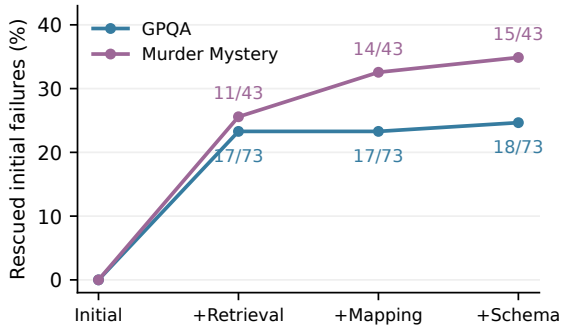



Figure 3: Cumulative oracle-assisted successes among Llama's initial failures, using denominators 73 and 43 throughout. Previously successful continuations are retained; the series is not an independent stage-effect comparison. Final oracle adaptation is excluded.

**Correct answers can omit the explicit schema path.** Across 400 natural model–item outcomes (two models on the same 200 questions), 23 final adaptation inputs have no schema text, roles, relations, conditions, inference rule, or exceptions. Twelve of those outcomes are correct: three Llama GPQA and one Llama murder outcome, and three DeepSeek GPQA and five DeepSeek murder outcomes. This inspects the actual recorded A input rather than an analogy-mention screen. A correct answer therefore does not guarantee that the designed transfer path delivered schema content. The check does not exclude spontaneous internal analogy or validate the schemas in the other paths.

**An initial stage distribution is incomplete.** The primary Llama cohort contains 73 initial GPQA failures and 43 murder failures. All GPQA failures receive initial R assignments in the retained diagnosis. Murder assignments are R in 19 cases, M in three, and A in 21. No initial S assignment appears in either cohort. Those labels describe the archived inspection order and criteria; their frequencies are not independently validated population estimates.

**Upstream correction does not ensure completion.** After the R replacement pass, 17 GPQA failures and 11 murder failures have successful continuations. After cumulative M replacement, the counts are 17 and 14; after S replacement, 18 and 15 (Figure 3). At that point 55 GPQA and 28 murder cases remain assigned to A: 75.34% and 65.12% of the original failure cohorts. The full state accounting appears in Appendix B.

The first GPQA assignments suggest attention to source construction. The corrected paths add a different observation: even with oracle-informed upstream artifacts, most initially unsuccessful Llama paths remain unsuccessful at adaptation. This is a conditional continuation outcome, not proof that adaptation was the hidden original cause. Conversely, the absence of initial S labels does not show that all schemas were adequate; S assignments appear after earlier corrections. A single initial label loses these distinctions.

The totals 45/100 GPQA and 72/100 murder combine natural successes with upstream-assisted successes. They must not be described as autonomous staged accuracy. Corrected final A outputs are excluded entirely. Because corrections accumulate, overlapping stage burdens also cannot be summed as numbers of distinct failed questions.

**A recoverable schema case illustrates the distinction.** In a saved DeepSeek murder case (item 87, question 0), the original schema has empty text, roles, relations, conditions, and inference rule, with `sufficiency=insufficient`. Adaptation reports that no schema is available, reasons from the target independently, and answers A while the benchmark key is B. A saved S-replacement continuation retains earlier stages, supplies an oracle-authored schema that distinguishes method-specific evidence from generic motive/access, and the same base model answers B. This example shows why an intermediate failure is more informative than either answer correctness or a verbal claim of analogy use. It is a documented assisted case, not an independent necessity test for S or a completed replication of the full cohort diagnosis.

## 5 Discussion

**Different measurements support different claims.** The studies form an evidence ladder rather than a single ranking. Correctness transitions describe practical changes. Explanation and random-source controls narrow their attribution. The factorial contrast provides evidence that answer benefits depend on designed correspondence. Explicit artifacts then expose content deficiencies and conditional continuation outcomes. No one layer establishes all the others, and evidence of supplied-source use does not automatically validate a self-generated staged path.

**What the framework adds.** The framework turns a general recommendation to "inspect reasoning" into specific questions: does the source

contain the necessary relation; does the mapping retain it; does the schema preserve applicability conditions; and does adaptation satisfy target constraints? These questions can guide source curation, prompt inspection, and error analysis. The empty-schema successes give an objective example: accuracy can credit an answer even when the designed path delivers no schema content. The remaining stage assignments require semantic inspection, while the recorded contracts and this absence check are mechanically recoverable. Their practical value is the distinction between available information and successful use of it.

**What remains unmeasured.** Stage-label agreement, blind reference validation, matched rephrasing controls, unchanged-artifact reruns, and information-access ablations remain outstanding. Consequently, we present a diagnostic protocol with descriptive case-study evidence, not a validated automatic causal attribution system. The all-R GPQA pattern particularly merits separating factual source errors from relational inadequacy and inspection bias. Restricting middle-stage inputs may itself contribute to failure. These questions can refine the framework; the present study does not resolve them by treating the stage outputs as internal cognition.

## 6 Related Work

**Analogy and theory.** ARN and AnaloBench evaluate identification of narrative correspondences (Sourati et al., 2024; Ye et al., 2024). Recognizing an analogue does not establish adaptation of its solution. Petersen et al. (2026) connect cognitive theories of analogy with NLP, including multiple processes beyond mapping. We build on these distinctions rather than claiming a new cognitive decomposition. Our contribution is an operational source-comparison-to-schema-to-target route with recorded interfaces and cumulative reference-assisted continuations.

**Exemplars and outcome attribution.** Analogical prompting generates relevant exemplars before solving a target (Yasunaga et al., 2024); SCUA studies scientific explanations through teacher analogies (Yuan et al., 2024). Qin et al. (2025) show that random examples can perform comparably, making relevance-based interpretations uncertain. Our controls extend the descriptive analysis to paired improvements and regressions. The factorial source-answer experiment complements this concern with positive evidence of selective answer benefit. Its source corpus is answer-informed, and correspondence is designed rather than independently established for every item.

**Intervening on reasoning traces.** Lanham et al. (2023) test how answers respond to changes in CoT, distinguishing stated reasoning from model behavior. Our replacements similarly test behavioral response to explicit text, but intervene at theory-motivated analogy interfaces and preserve a cumulative correction record. The general idea of intervention is not new. The particular contracts specify what content is available for comparison, abstraction, and target application, while our accounting excludes oracle final answers from solver success.

## 7 Conclusion

An analogy's evaluation should distinguish outcome changes, selective source benefit, and deficiencies in an explicit transfer attempt. We introduce a theory-grounded framework that makes source construction, mapping, schema induction, and adaptation inspectable through restricted artifacts. Controlled outcomes and a factorial source-answer study motivate this distinction; recorded continuations show that first-stage assignments do not describe the full corrected path. The framework supports a more precise diagnostic account while keeping natural performance, oracle-assisted continuation, and claims about hidden reasoning separate.

## Limitations

Our findings concern English multiple-choice scientific and narrative tasks. The narrative transfer cohort is the 103 completed items available during execution, not a documented representative sample. Scientific assistance conditions differ in content, instructions, and nominal word budgets; they are not a pure manipulation of analogy. Gemma's retained question counts differ by condition. Llama's final scientific set combines an initial provider run with recovery of failed requests through another provider. Family, size, quantization, and endpoint differences cannot be cleanly separated.

The target-story resampling intervals are conditional descriptions under a fixed corpus, source assignment, and prompt. They do not capture regeneration of sources, variation across random permutations, annotation uncertainty, or all cross-question dependencies introduced by reusing source material across cells. We do not use them for multiplicity-adjusted hypothesis tests or universal scaling claims. The valid-output subset is selected after generation. Answer-position strata compare different items and do not control their difficulty. Oracle-aware source generation and incomplete structural validation limit the interpretation of selective benefits.

The primary stage trajectories cover one base model on two 100-item cohorts and are retained aggregate reports. Natural direct/staged outcomes for a second model and one recoverable schema case broaden context, but do not establish a fully validated second-model replication of stage frequencies. Human review covers 30% of primary-study questions and outputs, with no inter-annotator agreement estimate. Thus inspection labels should be read as case-study assignments, not ground-truth causal mechanisms. The observable artifacts may be post-hoc products of target reasoning; neither structured output nor self-reported use establishes faithfulness.

Reference authors know the target key. References may indirectly encode answer guidance even without explicit disclosure, and correction changes content, style, and potentially length. Cumulative replacement and fresh downstream sampling do not identify independent stage effects. Unchanged-artifact repeats, matched rephrasing controls, blind reference audits, and interface ablations are not reported. Final adaptation replacement supplies an answer, so it is excluded from base-model continuation success. Direct answering and the staged workflow are not compute-matched; no efficiency or state-of-the-art claim is made.

## Ethical Considerations

We use existing scientific and synthetic narrative benchmarks and generated sources. A persuasive analogy can omit a decisive constraint and damage an otherwise correct answer, so benchmark improvements alone do not justify scientific or educational deployment. Dataset and model use remain subject to their licenses and service terms. Generative AI assisted the English manuscript, LaTeX, and offline analysis implementation; authors remain responsible for verifying text, sources, numerical provenance, and analysis. Experimental model use for source generation and oracle-authored references is described in the method. No new model calls or human annotations were added during this manuscript revision.

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# A Execution and Evaluation Details

This appendix specifies condition construction and accounting. The accompanying analysis inputs contain de-identified outcome metadata and retained aggregates. They do not constitute a new annotation of intermediate correctness. Archived experiments and the offline analyses in this manuscript are distinct: no model calls were added for this revision.

## A.1 Scientific conditions and complete accuracies

Condition 1 is the main-concept 300-word teacher analogy; 2 is one 600-word analogy; 3 and 4 are two 300-word and three 200-word examples. Conditions 5 and 6 are 300/600-word explanations. Conditions 7 and 8 are random same-domain/other-domain analogies. Condition 20 requests CoT; 21 requests solver-generated analogy with a 600-word generation ceiling. The broader archive includes an unrestricted-concept condition 0, which is not the selected analogy here.

Gemma response counts for conditions 1, 2, 3, 4, 5, 6, 7, 8, 20, and 21 are respectively 445, 444, 442, 441, 441, 444, 441, 444, 435, and 447. Correctness transitions use shared identifiers; a missing record differs from a retained response with no valid answer. Llama's final scientific set combines DeepInfra output with AvalAI recovery of failed requests. All 448 items remain in its accuracy denominator.

At a nominal 600-word total, Mistral's one-, two-, and three-example accuracies are 28.79%, 30.58%, and 26.56%; explicit mentions are 12.05%, 19.42%, and 29.24%. Thus additional references in output do not independently validate schema induction. The mention screen is lexical and includes rejecting an example.

## A.2 Narrative source construction

Sources contain narrative, question, choices, and answer fields. Generation preserves choice count, requests one correct source option, and caps source narrative length below 1,000 words. Source-choice reordering uses seed 20260901 and random assignment uses seed 20260903; the source key is updated after reordering. Withholding deletes only

| Condition | Llama-8B | Qwen-8B | Mistral-7B | Gemma-31B |
|---|---|---|---|---|
| Analogy, 300 words | 30.80 | 36.61 | 27.01 | 65.39 |
| Analogy, 600 words | 31.25 | 36.83 | 28.79 | 68.69 |
| Two analogies, $2 \times 300$ words | 31.92 | 35.27 | 30.58 | 69.00 |
| Three analogies, $3 \times 200$ words | 31.03 | 39.73 | 26.56 | 69.61 |
| Non-analogical explanation, 300 words | 31.03 | 37.72 | 27.68 | 68.25 |
| Non-analogical explanation, 600 words | 30.58 | 36.61 | 25.89 | 68.47 |
| Random same-domain analogy, 300 words | 31.25 | 36.61 | 27.68 | 64.85 |
| Random other-domain analogy, 300 words | 29.69 | 39.73 | 28.13 | 67.12 |
| Chain-of-thought | 32.14 | 42.19 | 16.96 | 66.52 |
| Self-analogy | 28.79 | 43.30 | 22.54 | 64.06 |

Table 4: Full scientific accuracies (%). Llama, Qwen, and Mistral retain all 448 question responses per condition. Gemma has differing retained question sets, so small column differences can include composition effects.

the answer field. An earlier condition that also removed source choices is excluded from the factorial comparison.

Corresponding/random narrative lengths average 371.41/371.41 words for murder, 189.14/228.76 for objects, and 296.57/256.95 for teams. Both full-corpus means are 318.99 words because random assignment uses the same source set. Equality at corpus level does not imply item- or task-level length matching. The four same-story controls are recorded in the assignment metadata and excluded together from all cells in the sensitivity analysis.

Small Llama models use Cloudflare, Llama-8B uses Novita FP8, and Llama-70B uses DeepInfra; Qwen uses DeepInfra FP8. These deployments do not isolate model size from infrastructure. Output extraction status is retained separately from correctness.

### A.3 Conditional uncertainty and sensitivity

For item $i$, average correctness over saved runs to obtain $\bar{y}_{ic}$ for cell $c$. Its contrast is $d_i = \bar{y}_{iB} - \bar{y}_{iA} - \bar{y}_{iD} + \bar{y}_{iC}$. The reported interaction is $100 \sum_i d_i/n$. Group identifiers combine task and story ID. Each bootstrap draw samples 88 groups with replacement, retaining every question in a selected group and all its cells/runs. The draw's statistic divides by its number of sampled questions, which may vary with cluster size. The percentile interval uses 20,000 draws and seed 20261012.

This procedure holds constructed sources and the permutation fixed. Source material appears in different cells across questions; resampling target groups does not address every dependency or all possible new assignments. The intervals describe conditional resampling stability, not a confirmatory causal test. No adjustment for exploring six model interactions is applied.

In $B$, 51 questions have matching source/target answer indices and 52 differ. Three Llama runs yield 153/156 events, not that many independent questions. Correct matching/differing counts are 135/1 for 1B, 93/83 for 3B, 114/89 for 8B, and 121/102 for 70B. The differing strata may differ in difficulty; these counts are not a randomized reordering experiment.

## B Diagnostic Accounting and Recoverable Interfaces

The overlapping stage burdens are R/M/S/A = 73/26/15/55 for GPQA and 19/6/1/28 for murder. Summing these would count paths more than once. These values come from retained aggregate diagnostics. Some restored graphics multiply observed counts by two; the present analysis returns to 100 questions per task and never treats scaled counts as additional observations. Older 50-item reports are not mixed with the 100-item results.

The Llama identifier is `cf.llama-3.1-8b-instruct-fast`; the contextual second model is `deepseek-v4-flash`. Natural direct/staged transitions (incorrect-to-correct / correct-to-incorrect) are 15/13 and 10/16 for Llama GPQA/murder, and 19/14 and 11/12 for DeepSeek. These are question-matched observed outputs, not a token-matched architecture comparison.

A recoverable individual DeepSeek case is `musr_murder_mystery:87:0`. The saved continuation is marked completed and oracle-assisted, dated 2026-10-02. Its chain records original base-model R and M, a strong manual S replacement, and base-model A rerun. The schema reference contrasts method-specific evidence with generic motive/access. The extract preserves the empty original schema, the initial unresolved schema check, the replacement, provenance, and answer

| Model | Invalid $A$ | Invalid $B$ | Invalid $C$ | Invalid $D$ | Valid-all $n$ | Valid-all $\Delta$ | Exclude same-story $\Delta$ |
|---|---|---|---|---|---|---|---|
| Llama-1B | 24 | 14 | 29 | 4 | 80 | 7.92 | $-0.34$ |
| Llama-3B | 2 | 4 | 0 | 5 | 98 | $-0.34$ | $-1.35$ |
| Llama-8B | 0 | 1 | 0 | 0 | 102 | 14.05 | 13.80 |
| Llama-70B | 13 | 9 | 8 | 7 | 88 | 14.02 | 13.13 |
| Qwen-14B | 24 | 23 | 40 | 39 | 58 | 12.07 | 9.60 |
| Qwen-32B | 13 | 19 | 20 | 25 | 65 | 10.00 | 12.63 |

Table 5: Sensitivity diagnostics. Invalid counts are response events, out of 309 per Llama cell or 206 per Qwen cell. Valid-all keeps questions valid in every cell and execution, a post-output selected subset. Excluding same-story controls uses the same 99 questions in all cells. Interactions are percentage points.

| State | Success | R | M | S | A |
|---|---|---|---|---|---|
| GPQA: 73 initial Llama errors | | | | | |
| Initial | 0 | 73 | 0 | 0 | 0 |
| After R | 17 | 0 | 26 | 7 | 23 |
| After M | 17 | 0 | 0 | 15 | 41 |
| After S | 18 | 0 | 0 | 0 | 55 |
| Murder: 43 initial Llama errors | | | | | |
| Initial | 0 | 19 | 3 | 0 | 21 |
| After R | 11 | 0 | 6 | 0 | 26 |
| After M | 14 | 0 | 0 | 1 | 28 |
| After S | 15 | 0 | 0 | 0 | 28 |

Table 6: Retained aggregate state accounting. Each row conserves its original error cohort. "After" denotes cumulative replacement; successful earlier continuations are carried forward. Final oracle adaptation is excluded.

A-to-B transition. The stored "strong manual replacement" label is a provenance label, not proof of independent human validation or verified authorship by a particular reference model.

### B.1 Schema-content absence

The absence check reads the schema supplied to the final recorded adaptation attempt, not merely the earlier S output or a self-report. It flags an input only when all six content fields are empty or absent: schema text, roles, relations, applicability conditions, solution/inference rule, and boundaries/exceptions. A sufficiency status alone is not schema content. Llama has 10 such GPQA inputs (three correct answers) and two murder inputs (one correct); DeepSeek has four (three correct) and seven (five correct). The total is 23 model–item outcomes, 12 correct. Models share 200 questions, so 400 outcomes are not 400 independent questions. The audit concerns the designed explicit information path, not unobserved reasoning.

### B.2 Recorded input-key audit

For the 400 natural staged cases in the matched Llama/DeepSeek cohorts, recorded prompt objects expose only the expected input keys: R receives `target, source_count`; M receives `sources`; S receives `mapping_package`; and A receives `target, schema`. The audit covers every recorded attempt and verifies that downstream input values match recorded parent outputs, with the target unchanged on re-entry. It finds four stage templates, all tagged `1.1-dev-6`. Among 400 workflows, 391 record four attempts, five record five, two record six, one records seven, and one records eight. These counts include retries and exclude oracle continuations.

This mechanical audit validates recorded explicit interfaces, not the correctness of outputs, absence of indirect target information in generated sources, or provider-side state. The recorded M task includes suspect/motive examples even for GPQA. We preserve this historical wording and report the possible domain bias rather than silently changing a prompt and attributing old outcomes to the new wording.

### B.3 Offline analysis package

The bundled outcome metadata, source-assignment IDs, stage templates, aggregate states, and compact case extract support reproducing the new offline statistics and both figures without model calls. `scripts/analyze_retained_results.py` checks cohort sizes, matching identifiers, four-cell coverage, and the permutation; `scripts/build_figures.py` uses bundled counts. A rendering adapter accepts a caller-supplied stateless completion function, instantiates recorded contracts, and continues after artifact replacement while retaining upstream outputs. Replacing A leaves no solver continuation. The adapter has no built-in API connection and is not the historical retry implementation. The analysis manifest records input hashes. This package reproduces reported summaries, not the original live API experiments or a fresh semantic annotation of all intermediate artifacts.